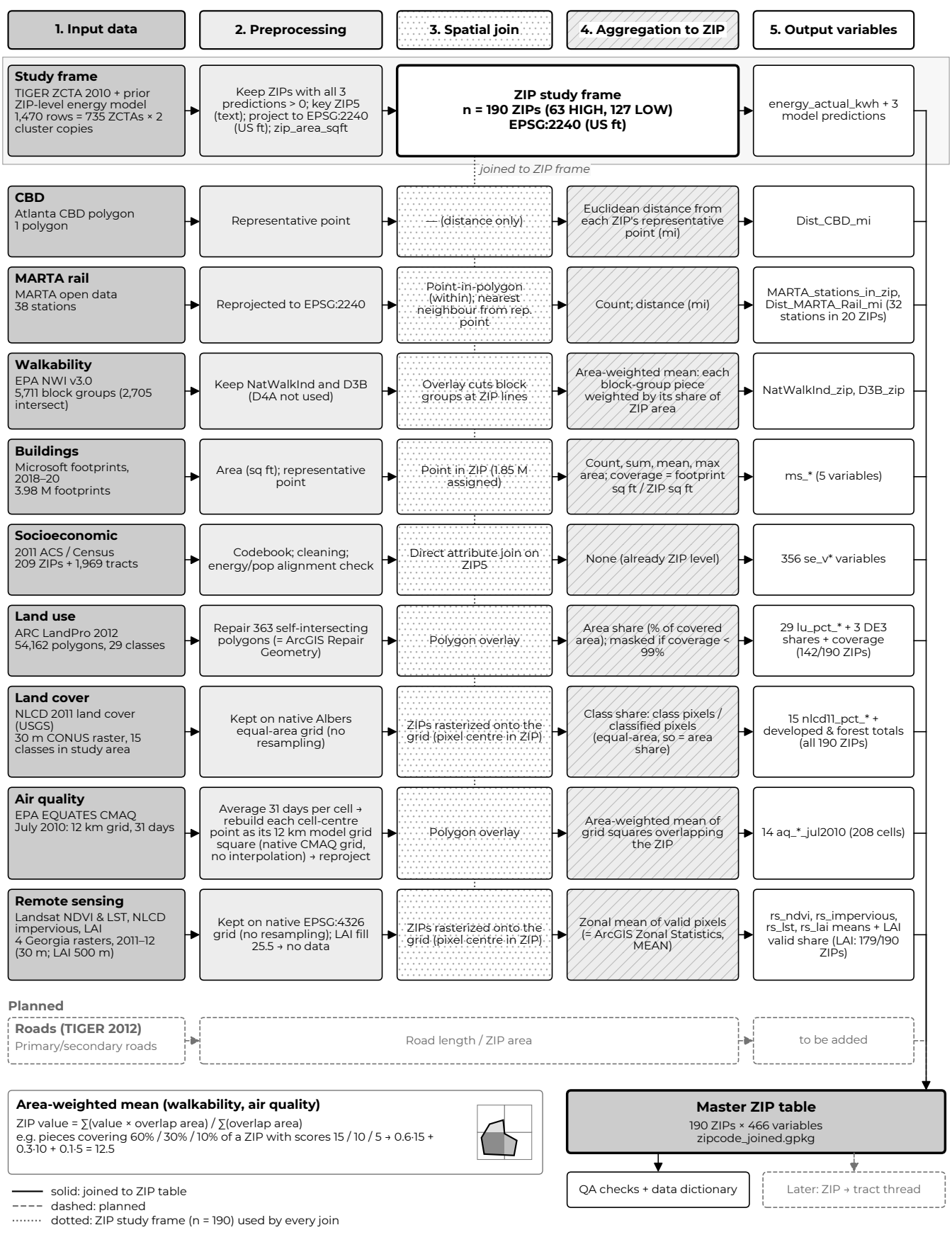

1. Input data
2. Preprocessing
3. Spatial join
4. Aggregation to ZIP
5. Output variables
Study frame
TIGER ZCTA 2010 + prior ZIP-level energy model
1,470 rows = 735 ZCTAs × 2 cluster copies
Keep ZIPs with all 3 predictions > 0; key ZIP5 (text); project to EPSG:2240 (US ft); zip_area_sqft
ZIP study frame
n = 190 ZIPs (63 HIGH, 127 LOW)
EPSG:2240 (US ft)
energy_actual_kwh + 3 model predictions
joined to ZIP frame
CBD
Atlanta CBD polygon
1 polygon
Representative point
— (distance only)
Euclidean distance from each ZIP's representative point (mi)
Dist_CBD_mi
MARTA rail
MARTA open data
38 stations
Reprojected to EPSG:2240
Point-in-polygon (within); nearest neighbour from rep. point
Count; distance (mi)
MARTA_stations_in_zip, Dist_MARTA_Rail_mi (32 stations in 20 ZIPs)
Walkability
EPA NWI v3.0
5,711 block groups (2,705 intersect)
Keep NatWalkInd and D3B (D4A not used)
Overlay cuts block groups at ZIP lines
Area-weighted mean: each block-group piece weighted by its share of ZIP area
NatWalkInd_zip, D3B_zip
Buildings
Microsoft footprints, 2018–20
3.98 M footprints
Area (sq ft); representative point
Point in ZIP (1.85 M assigned)
Count, sum, mean, max area; coverage = footprint sq ft / ZIP sq ft
ms_* (5 variables)
Socioeconomic
2011 ACS / Census
209 ZIPs + 1,969 tracts
Codebook; cleaning; energy/pop alignment check
Direct attribute join on ZIP5
None (already ZIP level)
356 se_v* variables
Land use
ARC LandPro 2012
54,162 polygons, 29 classes
Repair 363 self-intersecting polygons (= ArcGIS Repair Geometry)
Polygon overlay
Area share (% of covered area); masked if coverage < 99%
29 lu_pct_* + 3 DE3 shares + coverage (142/190 ZIPs)
Land cover
NLCD 2011 land cover (USGS)
30 m CONUS raster, 15 classes in study area
Kept on native Albers equal-area grid (no resampling)
ZIPs rasterized onto the grid (pixel centre in ZIP)
Class share: class pixels / classified pixels (equal-area, so = area share)
15 nlcd11_pct_* + developed & forest totals (all 190 ZIPs)
Air quality
EPA EQUATES CMAQ
July 2010: 12 km grid, 31 days
Average 31 days per cell → rebuild each cell-centre point as its 12 km model grid square (native CMAQ grid, no interpolation) → reproject
Polygon overlay
Area-weighted mean of grid squares overlapping the ZIP
14 aq_*_jul2010 (208 cells)
Remote sensing
Landsat NDVI & LST, NLCD impervious, LAI
4 Georgia rasters, 2011–12 (30 m; LAI 500 m)
Kept on native EPSG:4326 grid (no resampling); LAI fill 25.5 → no data
ZIPs rasterized onto the grid (pixel centre in ZIP)
Zonal mean of valid pixels (= ArcGIS Zonal Statistics, MEAN)
rs_ndvi, rs_impervious, rs_lst, rs_lai means + LAI valid share (LAI: 179/190 ZIPs)
Planned
Roads (TIGER 2012)
Primary/secondary roads
Road length / ZIP area
to be added
Area-weighted mean (walkability, air quality)
ZIP value = ∑(value × overlap area) / ∑(overlap area)
e.g. pieces covering 60% / 30% / 10% of a ZIP with scores 15 / 10 / 5 → 0.6·15 + 0.3·10 + 0.1·5 = 12.5
Master ZIP table
190 ZIPs × 466 variables
zipcode_joined.gpkg
QA checks + data dictionary
Later: ZIP → tract thread
solid: joined to ZIP table
dashed: planned
dotted: ZIP study frame (n = 190) used by every join